

Enterprise IT Infrastructure Plan: ABC Startup Solutions

Course: ITEP 414 – System Administration and Maintenance
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PART 1: Company Profile

- **Company Name:** ABC Startup Solutions
- **Nature of Business:** Software Development & Custom Tech Solutions
- **Company Vision:** To become a leading provider of innovative, scalable, and secure software applications that empower modern digital transformation.
- **Company Mission:** To engineer high-performing software tailored to enterprise client needs while maintaining data integrity, security, and exceptional user experience.
- **Office Location:** 4th Floor, Tech Hub Building, Santa Cruz, Laguna, Philippines

Organizational Structure

- **Chief Executive Officer (CEO)**
 - **Head of Technology** → IT Department (5 Employees)
 - **Head of Human Resources** → HR Department (4 Employees)
 - **Head of Finance** → Finance Department (5 Employees)
 - **Head of Sales** → Sales Department (6 Employees)

Employee Distribution

Department	Headcount	Key Responsibilities
Information Technology	5	Software engineering, dev ops, system maintenance, tech support
Human Resources	4	Recruitment, payroll administration, employee policy compliance
Finance	5	Financial reporting, budgeting, accounting, software billing
Sales	6	Client acquisition, product demos, account management
Total	20	

PART 2: Enterprise Hardware Inventory

Asset ID	Hardware Item	Quantity	Department	Purpose
HW-DESK-01	High-Performance Workstation (32GB RAM, i7/Ryzen 7)	5	IT	Required for local development environments, virtualization, and compiling code efficiently
HW-DESK-02	Mid-Range Desktop Computer (16GB RAM, i5/Ryzen 5)	9	HR (4), Finance (5)	Stationary, reliable setups for data entry, financial auditing, and HR information systems
HW-LAP-01	Portable Business Laptop (16GB RAM, Lightweight)	6	Sales	Allows mobility for client meetings, remote pitches, and dynamic work environments
HW-MON-01	27" Dual-Monitor Setups	14	IT (5), HR (4), Finance (5)	Increases productivity through multi-window productivity and side-by-side data comparison
HW-MON-02	24" Single Monitor (Docking setup)	6	Sales	Desk workstation connection points for laptops when sales reps are in-office
HW-SRV-01	Rackmount Server (Xeon/EPYC, 64GB ECC RAM, RAID-10)	1	Shared / IT Infrastructure	Hosts central active directory/file servers, local databases, and staging web apps
HW-NAS-01	4-Bay NAS Storage (16TB Raw RAID 5)	1	Shared / IT Infrastructure	Centralized, secure storage for local daily backups and file sharing
HW-BKP-01	4TB Rugged External Hard Drive	2	IT / System Admin	Offsite/cold backup rotation to comply with disaster recovery protocols
HW-PRN-01	Enterprise Multi-Function Network Laser Printer	1	Shared (Office Floor)	Centralized document scanning, printing, and legal documentation
HW-UPS-01	3000VA	1	Server Room /	Backup power for

	Line-Interactive Smart UPS		Rack	core network devices, server, and NAS during power fluctuations
HW-UPS-02	650VA Desktop UPS Unit	14	IT, HR, Finance Desktops	Prevents sudden shutdown and data loss on workstation computers

PART 3: Enterprise Software Inventory

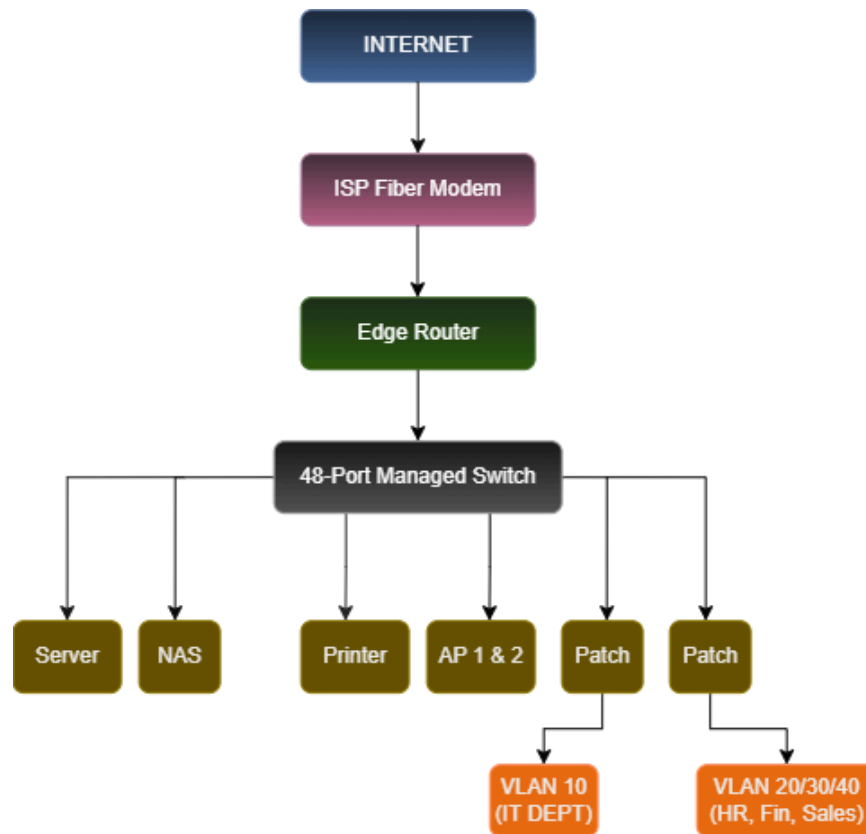
Software	Version	License Type	Purpose
Windows 11 Pro	22H2 / 23H2	Commercial Volume License	Standard desktop OS providing BitLocker encryption and domain join features
Ubuntu Server	24.04 LTS	Open-Source / Free	Robust server OS to host web services, database containers, and staging apps
Microsoft Office 365	Business Standard	Subscription (Per User)	Business communications, documentation, spreadsheets, and email host
VS Code	Latest Stable	Open-Source / Free	Primary integrated development environment (IDE) for software engineering
Git	Latest Stable	Open-Source / Free	Version control tool installed on development machines
Github Desktop	Latest Stable	Open-Source / Free	GUI interface for managing repositories and GitHub workflows
VirtualBox	7.0.x	Open-Source (GPLv2)	Virtualization software for testing applications across isolated OS environments
Google Chrome	Latest Stable	Freeware	Standard corporate web browser for accessing cloud portals and debugging
Microsoft Defender	Integrated / Enterprise	Included with OS / M365	Centralized antivirus, malware prevention, and host firewall management
AnyDesk	Enterprise Edition	Commercial License	Remote support

			software used by IT technicians for employee troubleshooting
7-Zip	Latest Stable	Open-Source (GNU LGPL)	High-compression utility for secure archive management and data packaging

PART 4: Enterprise Network Inventory

Item Description	Quantity	Purpose & Specifications
ISP Fiber Modem	1	Modulates optical internet signals from the Internet Service Provider into ethernet
Edge Router	1	High-throughput enterprise gateway handling routing tables and WAN connectivity
Hardware Firewall	1	Dedicated security appliance providing deep packet inspection, IPS, and VPN support
48-Port Managed Switch	1	Layer 2/3 Managed PoE+ switch connecting all desktops, servers, printers, and APs with VLAN tagging capabilities
Wireless Access Point (AP)	2	Ceiling-mounted Wi-Fi 6 APs providing secure corporate and guest wireless access
24-Port Patch Panel	2	Organized termination point for structured horizontal CAT6 network cables
CAT6 Ethernet Cable Spool	2 (1000ft)	High-speed gigabit cabling for horizontal runs across workstations and network devices
RJ45 Connectors & Boots	1 Box (100pcs)	Connectors used for custom patch cord creation and cable terminations

PART 5: Enterprise Network Diagram



PART 6: System Administration Roles

1. Helpdesk Technician

- **Responsibilities:** First point of contact for technical support; logs and tracks support tickets; provisions desktop hardware and peripheral setups; troubleshoots software/OS glitches
- **Skills:** User support, hardware diagnostics, active listening, Windows/macOS troubleshooting, basic networking
- **Common Tools:** Ticket management systems (Jira, Zendesk), remote support tools (AnyDesk, TeamViewer), Active Directory Users & Computers
- **Certifications:** CompTIA A+, Microsoft Certified: Modern Desktop Administrator Associate

2. Network Administrator

- **Responsibilities:** Configures and maintains routers, switches, firewalls, and APs; monitors network bandwidth and uptime; manages VLANs and subnets; enforces network perimeter security
- **Skills:** IP routing protocols, switching architecture, packet analysis, firewall rule management, VPN configuration.
- **Common Tools:** Wireshark, Cisco Packet Tracer, PuTTY, PRTG Network Monitor, SolarWinds
- **Certifications:** Cisco Certified Network Associate (CCNA), CompTIA Network+

3. Linux System Administrator

- **Responsibilities:** Deploys and maintains Linux servers; manages user permissions and system security; writes shell scripts for task automation; monitors server load, memory, and disk usage

- **Skills:** Bash scripting, Linux kernel parameters, systemd, SSH configuration, web server management (Nginx/Apache), package management
- **Common Tools:** Terminal, SSH, Htop/Top, Ansible, Cron, systemctl, Docker
- **Certifications:** Red Hat Certified System Administrator (RHCSA), Linux Professional Institute LPIC-1, CompTIA Linux+

4. Cloud Administrator

- **Responsibilities:** Provisions and configures cloud resources (VMs, storage buckets, virtual networks); monitors cloud infrastructure spending; implements cloud security and identity policies
- **Skills:** Cloud architecture design, Infrastructure as Code (IaC), identity access management (IAM), cloud networking, cost optimization
- **Common Tools:** AWS Management Console, Azure Portal, Terraform, AWS CLI, CloudWatch
- **Certifications:** AWS Certified Solutions Architect - Associate, Microsoft Certified: Azure Administrator Associate (AZ-104)

Within the ABC Startup Solutions Company, these four individuals provide a connected operational layer that allows for business continuity. If there is a problem, such as a connection issue by a developer to a cloud database, the helpdesk technician becomes the first point of contact who checks local hardware and submits documented notes. The network administrator makes sure that the VLANs and the firewall are configured locally. At the same time, the Linux system administrator takes care of making sure that the application server operating system and containers are functioning properly and not constrained by resources. Finally, the Cloud Administrator will make sure that cloud database access points and routing tables are set up properly.

PART 7: Infrastructure Recommendations

1. Internet Provider

- **Recommendation:** Enterprise Fiber Plan (e.g., PLDT Enterprise or Globe Business 300 Mbps Dedicated Fiber with 1:1 Committed Information Rate) with an optional secondary LTE/5G failover connection.
- **Justification:** Software development demands consistent upload and download speeds for git repositories and cloud deployments. A 1:1 CIR minimizes latency and guarantees operational uptime.

2. Server Specifications

- **Recommendation:** Dell PowerEdge R450 or equivalent Rackmount Server featuring Intel Xeon Silver processors, 64GB ECC DDR4 RAM, and 4x 1TB NVMe SSDs configured in RAID 10.
- **Justification:** ECC memory prevents enterprise data corruption, while RAID 10 offers both speed and drive redundancy, surviving a single disk failure without downtime.

3. Backup Strategy

- **Recommendation:** Implementation of the **3-2-1 Backup Rule**.
- **Justification:** Maintain 3 total copies of critical business data, on 2 different media types (Local NAS storage + Physical External Hard Drives), with 1 copy stored securely offsite (Encrypted Cloud Storage or physical rotation).

4. Security Recommendations

- **Antivirus / Endpoint Security:** Deploy **Microsoft Defender for Business / Endpoint**, enabling centralized policy management, automated threat containment, and real-time behavioral monitoring across all 20 devices.
- **Password Policy:** Mandatory minimum 14-character length, complex character enforcement, 90-day expiration cycle, multi-factor authentication (MFA) on all cloud/email logins, and integration of a business password manager (e.g., Bitwarden Enterprise).

5. Expansion Plan

- **Recommendation:** Deploy 48-port core switches instead of 24-port units and implement dynamic VLAN routing.
- **Justification:** ABC Startup Solutions currently utilizes ~25 active drops. Installing a 48-port switch allows double the employee headcount (up to 40+ staff) without replacing core networking hardware.

PART 8: Personal Reflection

As a result of working on this project, I have developed an understanding of the way the strategic design of information technology can fit in business activities. I understood that system administration is not just about unpacking the hardware, installing software, and configuring cables. Rather, the approach should be analytical, starting from the top and basing each technical decision on a certain organizational requirement. During the process of working on the project, I learned how to analyze processes in various departments, to design logically separated network topologies based on VLANs, to estimate hardware requirements for various employee job positions, and to implement various layers of security systems, including 3-2-1 backup system and centralized endpoint protection. Probably, the most difficult part of the project was developing the logical network topology, as it implied finding the right compromise between the strict isolation and efficiency of the network. In Draw.io, designing the network required much thought so as to provide the perfect isolation for sensitive departments, such as the Finance Department and the HR Department, while simultaneously ensuring that these departments would have easy access to all the essential local assets, such as the database server, NAS, and shared printers. Also, I needed to clearly depict visual pathways for four distinct departmental VLANs.

Proper planning before the deployment of information systems is vital for the operation in an enterprise as the wrong deployment may result in significant operational downtime, wasted capital, and security problems. In the real world, purchasing hardware without doing any capacity planning beforehand usually ends up in problems with performance or over-provisioning of hardware. Besides, configuring the network of the enterprise without appropriate initial segmentation will create a flat network, thus making the sensitive corporate data vulnerable to unauthorized access and malware spread. Having a well-documented blueprint makes sure that the hardware will be purchased efficiently, the network will be routined properly, and security policies will be applied from the beginning of work. Thus, the completion of this project has significantly improved my technical skills, strategic thinking, and the ability to generate professional technical documentation. Taking on the role of a Junior System Administrator, I had the opportunity to learn how to consider business limitations, organize the hardware and software inventory, and map network pathways. These skills, as well as the practice in documenting configurations in Markdown and GitHub, will definitely make me a more structured and efficient System Administrator in my future IT career.